

accessibility of the data available on the EPA's website. Jones and Berrens (2017) in a large-scale study of the western US looked at 146 AQS sites and 101 IMPROVE sites. Jones *et al.* (2016) used data from seven monitoring stations located in Albuquerque's metropolitan area from EPA AQS.

Some studies have used modelled data as opposed to monitoring data including Fann *et al.* (2018b) and Rappold *et al.* (2014) who used the Community Multiscale Air Quality model (a suite of programs from the EPA for conducting air quality monitoring simulations) and NOAA's Smoke Forecasting System (SFS) respectively to estimate emissions with and without wildland fires, for prescribed fires as well as agricultural fires.

Smoke event periods and smoke event counterfactual

In a second step, the smoke event period and an accurate location need to be determined to ensure that the extent of the estimated health impacts is correct. Studies must be careful not to overestimate the geographic extent in *BenMAP-CE*, thereby including too many people and vice versa. This component can be made as complex or as simple as required by the study.

Wind direction can be taken into account (Moeltner *et al.* 2013), or the use of smoke dispersion models, but it is also possible to use a simpler approach such as a maximum distance, e.g. an 80-km radius from the monitoring station (Jones *et al.* 2016). One must also consider the influence of several wildfires occurring in one region at the same time, which may lead to an overestimation of the impact of one specific fire.

Studies vary significantly in their approach: Jones and Berrens (2017) used the NOAA/NESDIS hazard mapping system in combination with the location of monitoring sites to look for intersection, i.e. where the visual smoke from satellite pictures correlated with elevated levels of the pollutant from monitoring stations. Specifically, they defined a smoke event day as daily average PM_{2.5} levels of more than the 99th percentile of daily average readings per monitor site over the previous 5 years. The same protocol was followed by Jones *et al.* (2016). Both studies describe this methodology as a conservative estimate as they are potentially missing many subtle smoke impacts. Rappold *et al.* (2014) examined the 2008 Evans Road fire in eastern North Carolina where cumulative exposure to PM_{2.5} over the study period exceeded 50 $\mu\text{g m}^{-3}$. Smoke exposure estimates were gained using the NOAA's SFS, which uses the BlueSky smoke modelling framework to forecast the smoke event periods and the smoke event counterfactual. So far, there is no consistent method used across studies, with many studies weighing up the trade-off between better representation of the smoke event period and the complexity and difficulty of the method.

Dose-response functions

The choice of the DR function plays an important role as it creates the explicit link between the smoke and health outcomes incidences. *BenMAP-CE*'s library contains ~40 DR functions, which are specific to urban air quality only; however, the software does have the capability to import other functions.

Results vary significantly between wildfire- and non-wildfire- (i.e. urban air quality) specific DR functions. Differences stem from chemical differences in smoke, the public's perceptions of health risks and associated averting behaviour (Liu *et al.*

2017). Also, urban air quality DR functions generally describe relationships for lower (PM) pollutant concentrations but long duration of exposure as opposed to wildfire-specific DR with high PM pollutant concentrations for a short period of exposure.

Wegesser *et al.* (2009) showed that wildfire 'PM was much more toxic to the lung on equal weight basis than PM collected from ambient air'. Related to this, Jones *et al.* (2016) found wildfire-specific DR functions produced considerably higher morbidity costs than those that used in urban air studies: they compared different DR functions and found COI for their analysis to be US\$74 000 using urban DR functions and US\$111 000 using wildfire DR functions, nearly 70% more. Emergency room asthma costs were a staggering 2535% higher and hospital admission for all respiratory costs were also 44% higher than using respective urban air quality DR functions.

BenMAP-CE studies using urban DR functions internal to *BenMAP-CE* were Douglass (2008), Rappold *et al.* (2014) and Jones *et al.* (2016). Studies that used wildfire-specific functions were Jones and Berrens (2017) and for some functions Jones *et al.* (2016). Jones and Berrens (2017) used six wildfire-specific DR functions from the health literature covering emergency room (ER) asthma visits and respiratory visits (based on Reid *et al.* 2016), hospital admissions for pneumonia, hospital admissions for all respiratory illnesses, hospital admissions for asthma from Delfino *et al.* (2009) and mortality functions from Johnston *et al.* (2012). The wildfire-specific DR functions used in Jones *et al.* (2016) were also from Delfino *et al.* (2009) and Resnick *et al.* (2015), which are specific to the western US. In a variation, Rappold *et al.* (2014) used a state-wide tracking tool to record daily ED visits and therefore incidence rates in North Carolina.

As Jones *et al.* (2016) state, the choice of DR function, 'is an unsettled methodological issue [...] which is likely to be confronted by researchers estimating the costs of a wildfire event'. The lack of internal wildfire-specific DR functions in *BenMAP-CE* is a shortcoming of the software. It requires analysts who choose to use wildfire-specific functions to engage with the relevant and specialised health literature to identify the appropriate information and enter it into *BenMAP-CE*. To help this process, Reid *et al.* (2016) analysed 103 wildfire-specific DR functions for their suitability and bias in a review of recent literature. Their review represents a good resource for future researchers looking for wildfire-specific DR functions.

Economic values

In a next step, *BenMAP-CE* uses the Aggregate, Pool and Value (APV) configuration to specify the geographic level at which the results are calculated, how the incidence results are combined or pooled and which economic value to assign.

This involves *BenMAP-CE* applying VSL, WTP or COI values for estimated incidence levels for various endpoints based on the chosen DR functions. For VSL, even though *BenMAP-CE* offers the central VSL value of US\$7.4 million (2006 dollars), this can be adjusted to account for inflation and changes in income (we expect VSL to increase with income). This leads to somewhat different VSL values even for studies that all use *BenMAP-CE*, as shown in Table 2.

For morbidity, although *BenMAP-CE* includes various estimates, certain endpoints only have inbuilt COI (based on actual treatment costs and wages lost) applicable to them. As with the

health impact functions, however, it is possible to import the WTP or COI values. Studies in this area are therefore split between those that use internal and external values in *BenMAP-CE*. For example, to value premature deaths, [Fann *et al.* \(2018b\)](#) used the VSL estimate internal to *BenMAP-CE* from the EPA as well as for hospital admissions from respiratory symptoms, whereas [Jones and Berrens \(2017\)](#) used their own derived WTP value of US\$141.24 to value morbidity impacts while the VSL from the EPA was used to value mortality. By using WTP or COI values external to *BenMAP-CE*, a more reliable final estimate can be made as these functions can be taken from literature that is more specific to the study's context.

Discussion

The literature on adverse health effects from wildfires during the past 10 years has been shaped by changes in wildfire occurrences but also by the availability of new tools for analysis. This is particularly true for the western United States: the increasing incidence of wildfires in this part of the world, related among other things to climate change and a growing urban–wildland interface, has heightened interest in this type of research, which is reflected by the growing number of increasingly sophisticated publications in the field. The availability of *BenMAP-CE* has made the analysis more accessible for researchers from different disciplines. The software is easy to access and non-technical, with manuals and example applications available online.

Nevertheless, the diversity in estimates at all levels, the consideration of single or multiple wildfires with differing geographic extents and varying population density make direct comparison of studies difficult.

Although mortality usually only accounts for a small percentage of total number of incidences in studies, the high VSL makes this impact account for most of the costs in studies that assess mortality impacts. In our review, we found that all studies applied the concept of VSL to determine the cost of premature mortality and all of the studies used EPA (or the equivalent organisation outside the United States) estimates from different regulatory contexts. Despite this, the range of values is considerable, which makes comparison tricky. We suggest that all studies apply the current central EPA estimate of the VSL of US\$9.38 (2019 dollars) and report lower and upper values that are study-appropriate (e.g. considering different age groups). This would at a minimum allow a comparison of the central VSL estimates for each study.

We found that the next largest health-related costs stem from working days lost and then respiratory morbidity endpoints relying on COI estimates. COI for morbidity are also highly diverse, in particular when comparing actual treatment cost and average cost per exposed person. Indeed, even treatment cost for similar endpoints may vary from hospital to hospital and certainly from country to country. The meta-analysis type approach taken in *BenMAP-CE* where a distribution based on several studies is derived may be well suited to accommodate geographic cost differences and increase comparability between studies.

Although there are so far only four studies that estimated (non-endpoint specific) WTP for avoiding health impacts from wildfire smoke, their results are encouraging as they all arrive at central estimates between US\$100 and 150 employing three

different approaches (DBM, CVM and SWB). In general, an increased use of WTP estimates for better decision-making would be advantageous and, as such, a more diverse range is needed across different demographic and socioeconomic groups as well as for specific endpoints in order to gain a greater understanding of the impacts of wildfire.

To increase validity of *BenMAP-CE* results for wildfires, the software should include wildfire WTP estimates. The literature so far provides four suitable WTP estimates that could be permanently included in *BenMAP-CE* ([Richardson *et al.* 2012](#); [Richardson *et al.* 2013](#); [Jones 2018](#)) and more could be added as the research advances.

In addition, it would be helpful if *BenMAP-CE* included more wildfire-specific features that still need to be entered or adjusted manually by the user. The calculation of economic health cost from wildfires (as opposed to urban air quality changes) requires identifying the relevant epidemiological literature for suitable DR functions and the appropriate economic literature for WTP or COI estimates. However, even with the appropriate DR functions, owing to the analysis being focused on relatively low but long-term pollutant exposure in the context of urban air quality, the output is calculated as changes over an entire year, which can bias results.

In addition, there exists no standardised methodology for estimating smoke event periods and areas and *BenMAP-CE* in its current form does not allow an easy estimation of a short-term smoke event period (from a few weeks to a few months, the usual duration of a fire).

By providing more wildfire-specific parameters in *BenMAP-CE*, the research of impact studies will become easier to carry out, more standardised and comparable in the future. This may allow us to better understand the overall trends in the economic costs of the adverse health impacts from wildfire smoke and make this field of research more accessible.

Assuming a sufficient number of studies on the economic health cost of wildfire smoke will be published in the next few years, a meta-analysis of those studies may be feasible ([Van Houtven 2008](#)). The resulting WTP values could be used for benefit transfer ([Johnston *et al.* 2015](#)), for wider application in the economic impact analysis of wildfires. Alternatively, more studies such as that of [Richardson *et al.* \(2013\)](#) that derive WTP:COI ratios may be applied for benefit transfers. Indeed, benefit transfer values may bridge the gap between complex original studies and the shortcomings of *BenMAP-CE* studies.

Conclusion

In this literature review, we have provided a systematic overview of studies on the economic health-related cost from wildfires since 2006. We showed that most studies have a common approach of estimating health cost by starting with determining air pollution from fires, estimating increased morbidity and mortality incidences due to the air pollution and finally attaching economic values to those additional incidences. However, comparisons between studies are limited by the many choices that need to be made in the analysis process and that lead to results that differ significantly between studies.

Although there is still great potential in deriving more consistent health cost from wildfires, it is clear, however, that health-related cost from wildfires are always substantial and

often proportionally the highest among all types of costs from wildfires ahead of suppression cost: average yearly federal suppression cost between 2006 and 2018 amounted to US\$1.78 billion (National Interagency Fire Center 2020). Fann *et al.* (2018a) found the yearly health cost of short-term exposure to wildfire smoke across the USA to be between US\$13 and 24 billion and between US\$90 and 154 billion per year for long-term exposure between 2008 and 2012. This certainly suggests that from a cost–benefit point of view, the cost of suppression spending is justified by the potential (further) damage avoided from wildfire smoke. However, the better alternative to accepting high suppression cost may be to think about interventions (masks, air cleaners) that may decrease the health costs from wildfire.

This is where research can also contribute. As far as we know, thus far, there are no economic assessments of health costs of prescribed fires, which would constitute an important component of the cost–benefit analysis of fire management regimes. In addition, tools like *BenMAP-CE* allow us to go beyond impacts by analysing the benefits (avoided damage), for example of increasing awareness among the public about the dangers of wildfire smoke and putting in place interventions to reduce smoke impacts. Studies like Rappold *et al.* (2014) and Parthum *et al.* (2017) have already begun to make headway here and we believe that future research should focus, in addition to improving impact estimates, on the economic benefits and costs of mitigating the effects of the high indirect costs from smoke as demonstrated by the studies analysed in the present literature review.

Conflicts of interest

The authors declare no conflicts of interest.

Acknowledgements

We acknowledge the funding support of the USDA Forest Service Pacific Northwest Research Station under agreement no. 18-CR-11261955–017 for this research.

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